Paper Manager pack — Diffusion Policy (Sep 21)

Paper: Diffusion Policy: Visuomotor Policy Learning via Action Diffusion

arXiv: https://arxiv.org/abs/2303.04137 · RSS 2023 / IJRR extended

Site/code: https://diffusion-policy.cs.columbia.edu

PDF: /workspace/papers/diffusion-policy-2303.04137.pdf

Session: Sep 21 · due 9:30 AM · **You = Archaeologist (seat 1)**

Type: VLA Architectures

Sources: Summarizer + Senior Research (`/workspace/diffusion-policy-quiet/DELIVERABLE.md`)

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1. Paper summary

Conditional **DDPM in action space**: given observation `O_t`, denoise noise → action chunk `A_t` of prediction horizon `T_p`; execute `T_a` then **replan** (receding-horizon). Observations condition once (FiLM / cross-attn). Beats BC-RNN / IBC / BET by **+46.9%** avg relative across **15** tasks.

Wins: multimodal `p(A|O)`, high-dim action sequences, stable MSE training, synergy with **position** control. Journal adds limiting-case control analysis, vision ablations, 3 bimanual real tasks.

Club line to BEHAVIOR: DP chunk+replan → ACT → π_0/$\pi_{0.5}$ **flow matching** chunks → RTC soft-inpaint → BEHAVIOR winners (exec26/inpaint4 + correlated noise). Denoiser changed; **chunk–commit–replan–overlap** habit stayed.

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2. Keywords

Diffusion Policy, DDPM/DDIM, action chunking, receding-horizon, multimodal BC, FiLM, CNN vs Transformer, position vs velocity, flow matching (downstream), RTC, OpenPI/π_0

Default horizons (real Push-T style): `T_o≈2`, `T_p≈16`, `T_a≈8` (sometimes 6); DDIM 100→10–16 steps for real-time.

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3. Motion / control spine (priority)

Idea	Detail
Action chunking	Joint `T_p×d_a` sequence → temporal consistency + idle robustness
Receding-horizon	Exec `T_a < T_p`, then replan; warm-start optional
Sweet spot	`T_a` too long = sluggish; `T_a=1` = mode flicker; paper ~8
Multimodal	Langevin basins (Push-T L/R, Kitchen order)
NOT	Classical motion planner / Diffuser full-trajectory planning

→ **Flow / BEHAVIOR (CLEAR):** π_0 cites Chi'23, swaps DDPM→flow on PaliGemma; RTC soft-inpaints async chunks; BEHAVIOR 1st: predict 30 / exec 26 / inpaint 4.

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4. Role packs

Stakeholder (18+3) — seat 2

Canonical continuous-action BC before VLAs. Value = **action head + control loop**, not semantics. Real-robot credible; DDIM keeps ~10 Hz. 2026 OpenPI/BEHAVIOR stacks inherit these control assumptions.

Scientific Reviewer (8+3) — seat 5

Strengths: Clear problem→method; 15-task sim+real+bimanual; ablations that matter (`T_a`, latency, pos/vel).

Actionable: shared action interface for all baselines; absolute + hard-task aggregates not only +46.9%; note robomimic init bug lore; mark as control-module not foundation VLM; seed-level last-10 tables.

Verdict: Still the right cite for continuous action diffusion/flow + chunks.

Empiricist (11+3) — seat 4

H-horizon: receding `T_a≈8` balances consistency vs reactivity

H-DDIM: quality saturates ~10–16 steps

H-pos: position > velocity for DP

H-loop: open-loop long chunks fail under obs shift (same failure mode as long VLA chunks)

Priority	Experiment
P0	`T_a∈{1,2,4,8,16}` on Push-T/Lift state; closed-loop vs open-loop

P1	DDIM steps {4..100} infer-only; position vs velocity
P2	CNN vs Transformer; warm-start
Defer	Vision FT, Transport, full 15-task

Budget: scout ≤~50 GPU-h; state Push-T first ~4–12 GPU-h. Env: `/workspace/diffusion-policy-quiet/env_outline/`.

Archaeologist (11+3) — **YOU (seat 1)** — HIGH PRIORITY

Influential priors

Prior	Role
DDPM / score matching / Langevin	Generative denoising backbone
DDIM / iDDPM	Fast sampling for real-time control
Diffuser	Trajectory diffusion foil — DP is **actions-only + O-conditioned**
IBC	Implicit BC foil (unstable / slow)
BC-RNN / Robomimic / BET	Multimodal BC baselines
FiLM / minGPT	Conditioning / Transformer pieces
Mayne RHC	Classical receding-horizon control ancestry
Relay Kitchen	Long-horizon multitask benchmark

Contemporaries: ACT (parallel CVAE chunking); other diffusion policies (Pearce/Reuss/IDQL).

CLEAR subsequent

Work	Link
π_0 (2410.24164)	Cites Chi'23; DDPM→**flow** on PaliGemma
OpenPI / $\pi_{0.5}$	Same chunked generative action habit
RTC (2506.07339)	Soft-inpaint async chunks
BEHAVIOR 1st (2512.06951)	30/26/4 + correlated noise; q~0.26
Comet (2512.10071)	Parallel RH / horizon ablations in challenge solutions

Talk arc (one slide): multimodal BC hacks → **DP locks generative chunks+replan** → VLAs swap flow for DDPM → RTC/BEHAVIOR soft-inpaint+correlated noise. Do **not** claim BEHAVIOR trains DP itself.

Visionary (11+3) — seat 3

2026 BEHAVIOR: control loop > bigger VLM alone for long-horizon fails. Bets: unify RTC+correlated inpaint; adaptive/event-triggered `T_a`; latency certificates; language-conditioned basin selection + DP-style commitment.

One-liner: DP is to continuous VLA motor control what early Transformers were to NLP sequences — the habit everyone still lives in after the backbone moved on. Skip Discord/Drive.

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5. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	Why action diffusion + real-robot credibility
Reviewer	8+3	Strengths + actionable cite hygiene
Empiricist	11+3	`T_a` grid + DDIM + closed vs open loop
Archaeologist	**11+3**	**Priors + DP→flow→RTC→BEHAVIOR timeline (HIGH)**
Visionary	11+3	2026 control-loop bets

Spine: chunk–commit–replan (`T_p`/`T_a`) as motion/control contract. Image-first; PDF for Mac.

Paths

Artifact	Path
This pack	`/workspace/diffusion-policy-quiet/KANTA_PACK.md`
Senior Research	`/workspace/diffusion-policy-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/diffusion-policy-2303.04137-summary.md`